



ग्रेटर नोएडा इंस्टीट्यूट ऑफ टेक्नोलॉजी (इंजीनियरिंग इंस्टीट्यूट)
GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)



Presentation
On
EMAIL AUTOMATION USING LLMS

Group Number: 26

Presented by

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Slide order follows the final research narrative exactly.

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Introduction

Email remains high-volume and latency-sensitive, yet users still handle most tasks manually.

Problem Context

- Email remains a high-volume, latency-sensitive communication channel, yet most users still perform reading, prioritization, and drafting manually.
- Existing assistance tools are often shallow, disconnected from the inbox workflow, or opaque in how they manage context and cost.
- This project proposes an inbox-native email intelligence system that performs thread understanding, summarization, reply synthesis, rewriting, classification, and urgency estimation.
- The system is designed around context preservation, preference-aware generation, and OpenRouter-backed model routing for efficient and explainable inference.

The screenshot displays a Gmail interface with an email thread. The thread consists of three messages from Vatsal Chaudhary to vatsal23m. The first message is titled 'Final project review and demo readiness' and asks for approval for a presentation order. The second message is a confirmation to review slides. The third message is a response stating that final screenshots and evaluation output will be prepared. An assistant panel is overlaid on the right side of the thread, providing a 'Generated Output' summary, action items, urgency, next steps, and recommended immediate action.

Final project review and demo readiness

Vatsal Chaudhary <vatsalchaudhary01@gmail.com> to vatsal23m

Please review the final slide deck, screenshots, evaluation summary, and demo sequence. We need approval for the presentation order, bac PM. Action items include confirming the architecture slide, validating the benchmark page, and preparing the fallback demo.

Vatsal Chaudhary to vatsal23m

Confirmed. I will review the slides and share the updated version by tomorrow.

Vatsal Chaudhary to vatsal23m

Great. I will prepare the final screenshots and evaluation output for the presentation.

Generated Output Copy Insert

Subject: Presentation Preparation - Urgent Review Required

Summary:
The email thread discusses final preparations for an upcoming presentation, with a critical deadline of Thursday at 5 PM. Key action items include:

Action Items:
1. Review and approve presentation slide deck
2. Confirm architecture slide details
3. Validate benchmark page
4. Prepare fallback demo sequence
5. Finalize presentation order and formatting

Urgency: High - Deadline is Thursday at 5 PM

Next Steps:
- Collaborators are actively working on slide deck, screenshots, and evaluation materials
- Expecting final review and comprehensive approval before the deadline

Recommended Immediate Action: Conduct a comprehensive review of all presentation materials to ensure alignment and readiness.

Gmail thread with the assistant panel visible for inbox-native analysis.

Research Objectives

The system is structured as a resilient orchestration layer, not a single-prompt demo.



Capture conversation context

Build an inbox-native interaction layer that captures active thread context without interrupting the user workflow.



Preserve memory and policy

Design a context-preserving orchestration layer with token budgeting, preference caching, and task-aware routing.



Personalize output

Incorporate tone control, signature reuse, and sender-specific response policy through memory-aware generation.



Expose telemetry

Track latency, token consumption, estimated cost, and routing efficiency for transparent evaluation.



Stay resilient

Support fallback inference, graceful degradation, and rate-limit-tolerant request handling.

Literature Survey

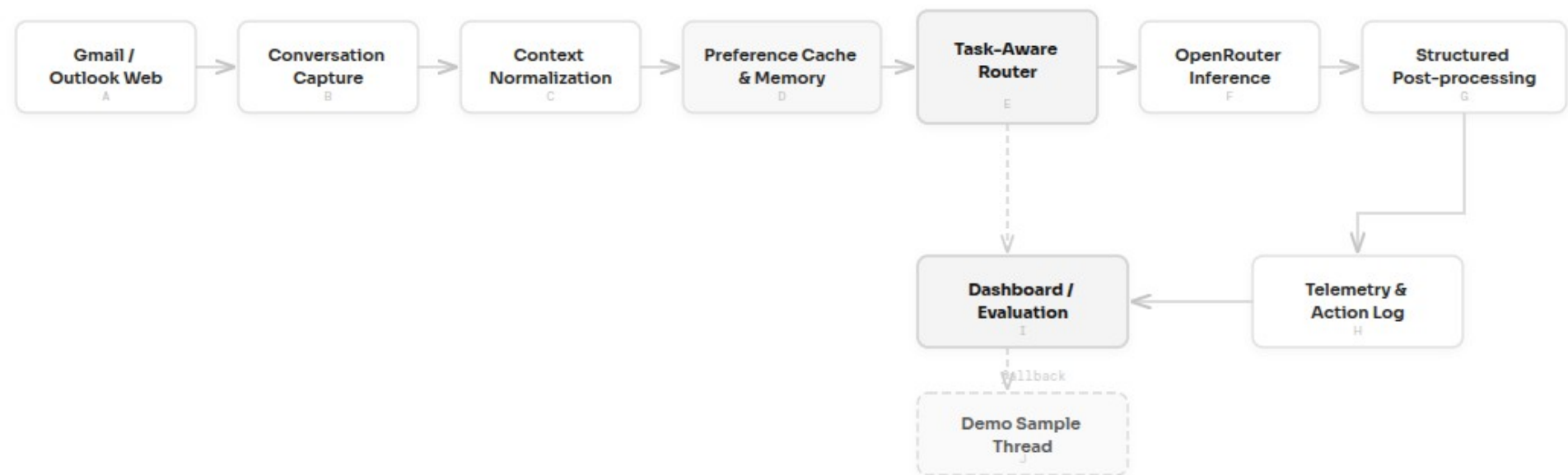
Existing approaches solve pieces of the workflow, but not the inbox-native orchestration problem end-to-end.

Smart compose systems	Fast short-form completion	Do not reason over full conversation context	Understands complete threads before generating assistance
Manual LLM use	Flexible prompting	Breaks workflow continuity and requires repeated context transfer	Operates directly inside Gmail and Outlook Web
Enterprise copilots	Strong integration	Often opaque in routing, personalization, and cost structure	Makes routing logic and telemetry visible
Template-based systems	Deterministic outputs	Lack semantic adaptability and tone control	Combines structure with context-aware generation

Research gap: a browser-native, context-preserving, cost-aware, and provider-flexible email orchestration system with explicit telemetry and evaluation.

Proposed Model

The architecture separates inbox capture, orchestration, provider routing, and telemetry.

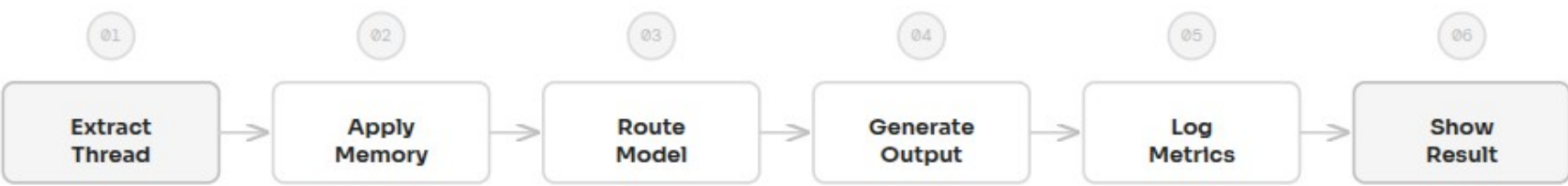


Provided diagram: presentation layer, orchestration layer, inference layer, and observability layer.

Inbox-native capture, context normalization, memory-aware routing, and telemetry together make the system explainable and cost-aware.

Discussion About Methodology & Algorithm

The processing pipeline keeps conversation context bounded, personalized, and observable from extraction through rendering.

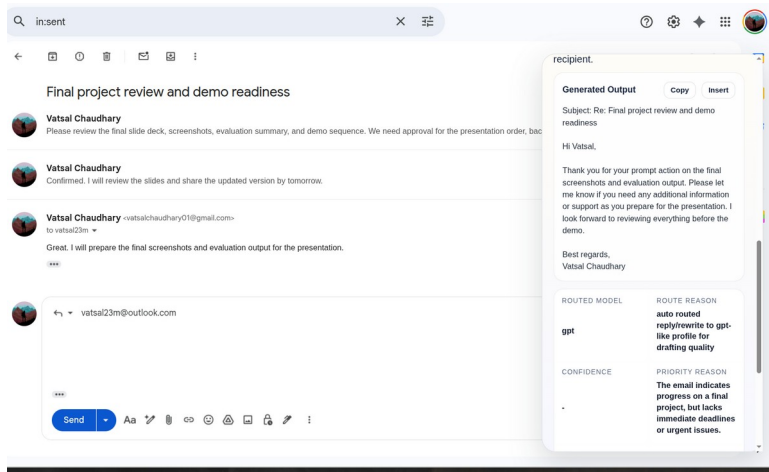


Provided pipeline diagram: Extract Thread -> Apply Memory -> Route Model -> Generate Output -> Log Metrics -> Show Result.

Algorithm

1. Extract the active thread and assemble bounded context.
2. Normalize the thread by removing UI noise.
3. Resolve user memory, tone, signature, and profile.
4. Route the task to the appropriate OpenRouter profile.
5. Generate the output with task-specific prompts.
6. Parse structured metadata such as confidence and action items.
7. Record telemetry including route reason, latency, token usage, and estimated cost.
8. Render the result and persist the action history.

Telemetry: route reason, latency, token usage, and estimated cost are captured for every run.



Extension result card used to render structured replies.

Conclusion & Future Work

The project turns email handling into an observable orchestration workflow rather than isolated text generation.

Conclusion

- The project shows that email automation becomes substantially more effective when generation is combined with thread context, model routing, and preference memory.
- The system shifts email handling from manual drafting to an intelligent orchestration workflow with visible cost and latency trade-offs.
- The architecture also improves explainability through route reason, structured metadata, and telemetry.

Key outcome: the inbox becomes a context-preserving workspace with visible routing and cost controls.

Future Work

- 1 Stronger Outlook parsing and layout stabilization
- 2 Attachment-aware summarization
- 3 Multilingual support
- 4 Persistent long-term memory storage
- 5 Controlled user studies and quality scoring

Research Paper Progress

Current implementation progress is visible through the inbox workflow, memory layer, and evaluation artifacts.

Problem definition
Done

Literature survey
Done

System implementation
Done

Evaluation tuning
In progress

Final writing
In progress

Final project review and demo readiness

Vatsal Chaudhary

to vatsal23m

Subject: Presentation Preparation - Urgent Review Required

Summary: The email thread discusses final preparations for an upcoming presentation, with a critical deadline of Thursday at 5 PM. Key action items include: 1. Review and approve presentation slide deck 2. Confirm architecture slide details 3. Validate benchmark page 4. Prepare fallback demo sequence 5. Finalize presentation order and formatting Urgency: High - Deadline is Thursday at 5 PM Next Steps: - Collaborators are actively working on slide deck, screenshots, and evaluation materials - Expecting final review and comprehensive approval before the deadline Recommended Immediate Action: Conduct a comprehensive review of all presentation materials to ensure alignment and readiness.

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Copy Insert

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Inbox analysis

Final project review and demo readiness

Vatsal Chaudhary

to vatsal23m

Subject: The Final project review and demo readiness

Hi Vatsal,

Thank you for your prompt action on the final screenshots and evaluation output. Please let me know if you need any additional information or support as you prepare for the presentation. I look forward to reviewing everything before the demo.

Best regards, Vatsal Chaudhary

Generated Output

Copy Insert

Subject: The Final project review and demo readiness

Hi Vatsal,

Thank you for your prompt action on the final screenshots and evaluation output. Please let me know if you need any additional information or support as you prepare for the presentation. I look forward to reviewing everything before the demo.

Best regards, Vatsal Chaudhary

Reply synthesis

MailPlus AI

MailPlus AI

Memory and Personalization

Set profile, tone, signature, and instructions used in replies

Current Memory Profile

Profile Name: Vatsal Chaudhary

Profile Type: Professional

Preferred Tone: Direct And Concise

Update Memory Settings

User Name: Vatsal Chaudhary

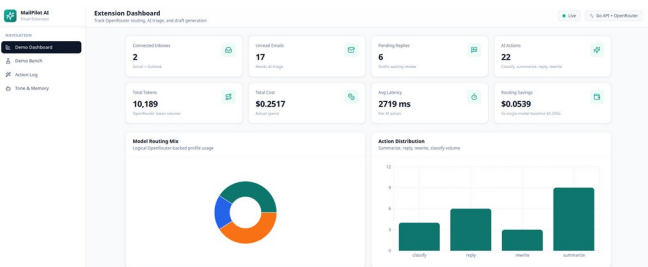
Preferred Tone: professional

Internal Tone: direct and concise

Memory and personalization

Current focus

Final evaluation numbers
Outlook stabilization
Paper polishing



Evaluation dashboard and compare mode

References

Selected sources used for the system design, routing approach, and inbox integration.

1. Vaswani, A. et al. Attention Is All You Need. NeurIPS, 2017.

2. Lewis, M. et al. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. ACL, 2020.

3. Google. Gmail API Documentation. <https://developers.google.com/gmail>

4. Microsoft. Microsoft Graph and Outlook Add-in Documentation. <https://learn.microsoft.com/graph>

5. OpenRouter. API Documentation and Model Routing Reference. <https://openrouter.ai/docs>